Smart21 ElectroLab

New Devices & ICs

Reference guide for the 34 parts added in versions 1.4–1.7 and a summary of the new features: what each one does, its pins, and how to use it.

What is included

Drivers & power ICs	Voltage regulator (7805 …), LM317 adjustable regulator, ULN2003 driver (7 × 500 mA), L293D motor driver (H-bridge), Optocoupler (PC817)
Power	Fuse, Transformer
Digital ICs	74HC00 / 08 / 32 / 86 quad gate, 74HC02 quad NOR, 74HC04 / 74HC14 hex inverter, 74HC74 dual D flip-flop, CD4017 decade counter, 74HC161 4-bit counter, CD4511 BCD to 7-segment decoder, 74HC138 3-to-8 decoder, 74HC165 parallel-in shift register
Analog ICs	74HC4051 analog multiplexer, 8-bit DAC (R-2R), 8-bit ADC (parallel), LM393 comparator (dual)
Semiconductors	Schottky diode (1N5819), SCR thyristor (2N5060)
Display & motion	Lamp (filament bulb), LED bar graph (10 LEDs), 8×8 LED matrix, I²C LCD (16×2 / 20×4), OLED 128×64 (SSD1306, I²C), Stepper motor (28BYJ-48)
Sensors	PIR motion sensor (HC-SR501), Temperature sensor (LM35), Temperature & humidity (DHT11 / DHT22), Joystick module
Switches & output	Keypad 4×4, DIP switch (4 positions)

New features in versions 1.6 and 1.7

New circuit: the New button opens a template chooser (Blank, Arduino project, Breadboard project, your own templates, Recent projects and every example as a template). Save as template: the toolbar button saves the current circuit; teachers can export and import template packs.

Arduino libraries: Stepper.h, Keypad.h, DHT.h, LiquidCrystal_I2C.h and Adafruit_SSD1306.h. Troubleshoot: eight circuits with one planted fault; “Check my fix” simulates the circuit and judges what it does. Datasheet: the ? button in the inspector. Logic analyzer and serial plotter tabs. Schematic view with standard symbols. Circuit report: picture, parts list, notes and sketch as a PDF.

How to use these parts

Drag a part from the left palette onto the workspace (the new section “Drivers & power ICs” holds the regulators and drivers). Select it to read its live values in the inspector, and hover a pin to see its name. Digital chips need real power: connect VCC and GND, and tie unused enable, reset and input pins to VCC or GND. A floating input reads LOW in the simulator.

Fourteen ready-made examples for these parts are in the examples gallery (7805 regulator, 555 + CD4017 running light, digital counter, stepper motor, keypad, LM393 night light, 555 one-shot, SCR latch, transformer, ADC voltmeter, DHT sensor, I²C LCD and OLED).

Drivers & power ICs

Voltage regulator (7805 …)

Turns a higher, wobbly DC voltage into a steady one. The 7805 gives 5 V as long as the input is at least ~7 V (the “dropout”). Put a capacitor (0.1–10 µF) at the input and output. It wastes the difference as heat: P = (Vin − Vout) × I, so 12 V → 5 V at 0.5 A is 3.5 W of heat. Limit about 1 A.

#	Pin / terminal
1	IN (input)
2	GND (common)
3	OUT (regulated output)

Inspector settings: Regulator

LM317 adjustable regulator

OUT stays exactly 1.25 V above ADJ. Put R1 (about 240 Ω) between OUT and ADJ, and R2 between ADJ and GND: Vout = 1.25 × (1 + R2 ÷ R1). A 5 kΩ potentiometer as R2 makes an adjustable 1.25–30 V supply. The input must be at least 3 V higher than the output. Needs about 10 mA minimum load (R1 provides it).

#	Pin / terminal
1	IN (input, ≥ Vout + 3 V)
2	ADJ (adjust)
3	OUT (output)

ULN2003 driver (7 × 500 mA)

Seven Darlington transistors in one chip: a small signal on INx (about 2 V or more) switches OUTx to GND and lets up to 500 mA flow through your load (relays, stepper motors, lamps, motors). The load goes between the supply + and OUTx. Connect GND to ground, and COM to the load supply + so the built-in flyback diodes protect against inductive spikes. Its Arduino-friendly inputs need no resistors.

Pins (first half)	Pins (second half)
1 IN1 input (HIGH ≈ 2 V or more switches OUT1 on)	16 OUT1 (open collector: sinks current to GND)
2 IN2 input (HIGH ≈ 2 V or more switches OUT2 on)	15 OUT2 (open collector: sinks current to GND)
3 IN3 input (HIGH ≈ 2 V or more switches OUT3 on)	14 OUT3 (open collector: sinks current to GND)
4 IN4 input (HIGH ≈ 2 V or more switches OUT4 on)	13 OUT4 (open collector: sinks current to GND)
5 IN5 input (HIGH ≈ 2 V or more switches OUT5 on)	12 OUT5 (open collector: sinks current to GND)
6 IN6 input (HIGH ≈ 2 V or more switches OUT6 on)	11 OUT6 (open collector: sinks current to GND)
7 IN7 input (HIGH ≈ 2 V or more switches OUT7 on)	10 OUT7 (open collector: sinks current to GND)
8 GND (emitters)	9 COM (flyback diodes: connect to the load supply +)

L293D motor driver (H-bridge)

Two H-bridges: it drives TWO DC motors in both directions (up to 600 mA each). Put a motor between 1Y and 2Y: 1A HIGH + 2A LOW spins it one way, the opposite spins it the other way, both equal = brake. Use the EN pin with analogWrite() to change speed. VCC (5 V) powers the logic and VS powers the motors (up to 36 V). Outputs reach about VS − 1.4 V; internal diodes protect against motor spikes. All four GND pins are joined.

Pins (first half)	Pins (second half)
1 EN1,2 enable channels 1 & 2 (PWM here for speed)	16 VCC1 logic supply (5 V)
2 1A input	15 4A input
3 1Y output (to motor)	14 4Y output (to motor)
4 GND	13 GND
5 GND	12 GND

6 2Y output (to motor)	11 3Y output (to motor)
7 2A input	10 3A input
8 VS motor supply (4.5–36 V)	9 EN3,4 enable channels 3 & 4

Optocoupler (PC817)

Passes a signal between two circuits with NO electrical connection: an LED inside shines on a phototransistor. Current through the LED (pins 1→2, needs a resistor!) lets the transistor conduct between COLLECTOR and EMITTER: the output current is about CTR × LED current. Use it to read mains-side or high-voltage signals safely, or to switch a load from a separate supply.

#	Pin / terminal
1	1 anode (LED +)
2	2 cathode (LED −)
3	4 collector
4	3 emitter

Inspector settings: Current transfer ratio

Power

Fuse

A deliberate weak link: if the current goes far above its rating the thin wire inside melts and opens the circuit before wires or parts burn. A fuse only blows once. Press Stop and Run to fit a new one. Place it in series with the supply, rated just above the normal current.

#	Pin / terminal
1	end A
2	end B

Inspector settings: Rating

Transformer

Two coils on one iron core. An AC voltage on the PRIMARY (left) appears on the SECONDARY (right) multiplied by the secondary ÷ primary turns: a 2 : 1 transformer halves the voltage, a 1 : 2 doubles it. It only works with AC (a steady DC gives nothing on the secondary). The two sides are NOT connected to each other, so connect one secondary pin to ground yourself. Dots mark the pins that swing together. Power out can never exceed power in: a step-up transformer gives less current.

#	Pin / terminal
1	primary 1 (dot)
2	primary 2
3	secondary 1 (dot)
4	secondary 2

Inspector settings: Turns ratio (primary : secondary), Primary inductance

Semiconductors

Schottky diode (1N5819)

A diode that drops only about 0.3 V instead of 0.7 V, and switches very fast. Used for reverse-polarity protection, power supplies with little waste, and as flyback diodes. Current flows anode → cathode (the bar).

#	Pin / terminal
1	anode
2	cathode (bar)

SCR thyristor (2N5060)

A diode you can switch ON with a pulse. A brief current into the GATE turns it on; it then keeps conducting from ANODE to CATHODE by itself, even after the gate pulse stops, until the current falls below about 5 mA (the “holding current”). Use it for latching alarms and power control. The gate needs about 0.3 mA or more, through a resistor.

#	Pin / terminal
1	anode (+)
2	cathode (−)
3	gate

Analog ICs

74HC4051 analog multiplexer

An 8-way analog switch: S0–S2 (S0 = 1, S1 = 2, S2 = 4) choose which of the channels Y0–Y7 is connected to COM through about 80 Ω. Signals pass in BOTH directions and can be analog voltages: read 8 sensors with one Arduino analog pin, or send one signal to 8 places. Tie /E and VEE to GND; the signals must stay between GND and VCC.

Pins (first half)	Pins (second half)
1 Y4 channel	16 VCC
2 Y6 channel	15 Y2 channel
3 COM (common in/out)	14 Y1 channel
4 Y7 channel	13 Y0 channel
5 Y5 channel	12 Y3 channel
6 /E enable (active LOW, tie to GND)	11 S0 select (value 1)
7 VEE (tie to GND)	10 S1 select (value 2)
8 GND	9 S2 select (value 4)

8-bit DAC (R-2R)

Digital-to-analog converter: turns an 8-bit number on D0–D7 (D7 = 128 … D0 = 1) into a voltage: OUT = VREF × number ÷ 256. Drive D0–D7 from 8 Arduino pins (or a 74HC595) and write a waveform with code. Connect VREF to 5 V for a 0–5 V range. The output is weak (a few mA): buffer it with an op-amp for loads.

Pins (first half)	Pins (second half)
1 D0 data bit (value 1)	7 D6 data bit (value 64)
2 D1 data bit (value 2)	8 D7 data bit (value 128)
3 D2 data bit (value 4)	16 VCC
4 D3 data bit (value 8)	15 VREF reference voltage (full scale)
5 D4 data bit (value 16)	14 OUT analog output
6 D5 data bit (value 32)	13 GND

8-bit ADC (parallel)

Analog-to-digital converter: measures the voltage on VIN and shows it as an 8-bit number on D0–D7: code = VIN ÷ VREF × 255 (0 V = 0, VREF = 255). Connect VREF to the supply (or leave it open to use VCC). In “continuous” mode the number follows the input all the time; in “START pulse” mode it only samples when START goes HIGH, and EOC (end of conversion) is LOW while START is HIGH. Read the 8 bits with an Arduino, or light 8 LEDs to see the binary number.

Pins (first half)	Pins (second half)
1 VIN analog input	14 VCC (4.5–6 V)
2 VREF reference (full scale); open = use VCC	13 D5 (32)

3 START (rising edge samples, in START-pulse mode)	12 D4 (16)
4 EOC end of conversion (HIGH = result ready)	11 D3 (8)
5 D7 (MSB, value 128)	10 D2 (4)
6 D6 (64)	9 D1 (2)
7 GND	8 D0 (LSB, value 1)

Inspector settings: Conversion

LM393 comparator (dual)

Two comparators. When IN+ is HIGHER than IN− the output transistor is OFF (the output floats: a pull-up resistor makes it HIGH); when IN+ is LOWER than IN− the output is pulled to GND. The outputs are “open collector”: always add a pull-up resistor (about 10 kΩ) from OUT to VCC. Perfect for light or temperature detectors. Add a resistor from OUT back to IN+ to make a Schmitt trigger.

#	Pin / terminal
1	1 OUT1 (open collector: needs a pull-up resistor)
2	2 IN1− (inverting input)
3	3 IN1+ (non-inverting input)
4	4 GND
5	8 VCC (2–36 V)
6	7 OUT2 (open collector: needs a pull-up resistor)
7	6 IN2− (inverting input)
8	5 IN2+ (non-inverting input)

Digital ICs

74HC00 / 08 / 32 / 86 quad gate

Four 2-input gates in one 14-pin chip. Pick the chip in the inspector: 74HC00 = NAND, 74HC08 = AND, 74HC32 = OR, 74HC86 = XOR. Each gate has inputs A, B and output Y. Inputs read HIGH above half the supply; unused inputs should be tied to GND or VCC (a floating input reads LOW here). Needs power on VCC and GND.

Pins (first half)	Pins (second half)
1 1A	14 VCC
2 1B	13 4B
3 1Y output	12 4A
4 2A	11 4Y output
5 2B	10 3B
6 2Y output	9 3A
7 GND	8 3Y output

Inspector settings: Chip

74HC02 quad NOR

Four 2-input gates in one 14-pin chip. Pick the chip in the inspector: 74HC02 = NOR. Each gate has inputs A, B and output Y. Inputs read HIGH above half the supply; unused inputs should be tied to GND or VCC (a floating input reads LOW here). Needs power on VCC and GND.

Pins (first half)	Pins (second half)
1 1Y output	14 VCC
2 1A	13 4Y output
3 1B	12 4B
4 2Y output	11 4A
5 2A	10 3Y output

6 2B	9 3B
7 GND	8 3A

74HC04 / 74HC14 hex inverter

Six NOT gates. The 74HC04 flips at half the supply. The 74HC14 has Schmitt-trigger inputs: it switches HIGH→LOW at about 65% of VCC and LOW→HIGH at about 35%, so slow or noisy signals give a clean output. Add one resistor and one capacitor to a 74HC14 and you have an oscillator. Unused inputs should be tied to GND or VCC.

Pins (first half)	Pins (second half)
1 1A	14 VCC
2 1Y output	13 6A
3 2A	12 6Y output
4 2Y output	11 5A
5 3A	10 5Y output
6 3Y output	9 4A
7 GND	8 4Y output

Inspector settings: Chip

74HC74 dual D flip-flop

Two flip-flops that remember one bit each. On the RISING edge of CLK the output Q copies the D input. /PRE (active LOW) forces Q = 1 and /CLR (active LOW) forces Q = 0 at any time: tie them to VCC when you do not use them, or the flip-flop is held in its reset state. Wire /Q back to D and the flip-flop divides the clock frequency by 2.

Pins (first half)	Pins (second half)
1 /1CLR (active LOW, tie to VCC)	14 VCC
2 1D data	13 /2CLR (active LOW, tie to VCC)
3 1CLK clock (rising edge)	12 2D data
4 /1PRE (active LOW, tie to VCC)	11 2CLK clock (rising edge)
5 1Q output	10 /2PRE (active LOW, tie to VCC)
6 /1Q inverted output	9 2Q output
7 GND	8 /2Q inverted output

CD4017 decade counter

A counter with 10 outputs: only ONE of Q0–Q9 is HIGH at a time, and it moves to the next output on every rising edge of CLK. RESET (HIGH) jumps back to Q0. INH (HIGH) freezes the count: tie it to GND. CO is HIGH for counts 0–4 and LOW for 5–9 (a clock ÷10). Pair it with a 555 for a running-light chaser! Outputs drive about 10 mA: use a resistor with every LED. Works from 3–15 V.

Pins (first half)	Pins (second half)
1 Q5	16 VDD (3–15 V)
2 Q1	15 RESET (HIGH = back to Q0)
3 Q0	14 CLOCK (rising edge)
4 Q2	13 CLOCK INHIBIT (HIGH = stop; tie to GND)
5 Q6	12 CARRY OUT (HIGH for counts 0–4)
6 Q7	11 Q9
7 Q3	10 Q4
8 GND	9 Q8

74HC161 4-bit counter

Counts 0–15 in binary on Q0–Q3 at every rising edge of CP. /MR (LOW) clears it to 0. /PE (LOW) loads P0–P3 on the next clock, so you can start from any number. Counting needs CEP and CET both HIGH:

tie them, and /MR and /PE, to VCC when unused. TC goes HIGH when the count is 15 (use it to chain counters). Connect Q0–Q3 to the CD4511 and you get a decimal display driver, or to LEDs to see binary.

Pins (first half)	Pins (second half)
1 /MR clear (active LOW, tie to VCC)	16 VCC
2 CP clock (rising edge)	15 TC terminal count (HIGH at 15)
3 P0 parallel data	14 Q0 (value 1)
4 P1 parallel data	13 Q1 (value 2)
5 P2 parallel data	12 Q2 (value 4)
6 P3 parallel data	11 Q3 (value 8)
7 CEP count enable (tie to VCC)	10 CET count enable (tie to VCC)
8 GND	9 /PE parallel load (active LOW, tie to VCC)

CD4511 BCD to 7-segment decoder

Turns a 4-bit number (A = 1, B = 2, C = 4, D = 8) into the segment pattern for a COMMON-CATHODE 7-segment display. Connect outputs a–g to the display segments (with resistors) and the display COM to GND. Numbers above 9 blank the display. /LT (LOW) lights every segment (lamp test), /BI (LOW) blanks the display, LE (HIGH) freezes the shown digit. Tie /LT and /BI to VDD and LE to GND for normal use. Works from 3–15 V.

Pins (first half)	Pins (second half)
1 B input (value 2)	16 VDD (3–15 V)
2 C input (value 4)	15 segment f
3 /LT lamp test (active LOW, tie to VDD)	14 segment g
4 /BI blanking (active LOW, tie to VDD)	13 segment a
5 LE latch enable (HIGH freezes, tie to GND)	12 segment b
6 D input (value 8)	11 segment c
7 A input (value 1)	10 segment d
8 VSS (GND)	9 segment e

74HC138 3-to-8 decoder

Three address inputs (A0 = 1, A1 = 2, A2 = 4) select ONE of eight outputs. The selected output goes LOW and all others stay HIGH (active-low outputs). The chip only works when /E1 and /E2 are LOW and E3 is HIGH: tie /E1 and /E2 to GND and E3 to VCC for always-on. Use it to choose one device out of many with only 3 Arduino pins.

Pins (first half)	Pins (second half)
1 A0 address bit (value 1)	16 VCC
2 A1 address bit (value 2)	15 /Y0 output
3 A2 address bit (value 4)	14 /Y1 output
4 /E1 enable (active LOW, tie to GND)	13 /Y2 output
5 /E2 enable (active LOW, tie to GND)	12 /Y3 output
6 E3 enable (active HIGH, tie to VCC)	11 /Y4 output
7 /Y7 output	10 /Y5 output
8 GND	9 /Y6 output

74HC165 parallel-in shift register

Reads 8 inputs with 2–3 Arduino pins (the opposite of the 74HC595). Pulse /PL LOW to capture D0–D7 into the register. Then, with /CE LOW, each rising edge of CP moves the data along and the next bit appears on Q7 (D7 first, D0 last). Read Q7 with digitalRead() after each pulse (or shiftIn()). Tie /CE to GND, DS to GND and /PL to VCC when not using them. Use it to read 8 buttons or the DIP switch.

Pins (first half)	Pins (second half)
1 /PL parallel load (LOW = capture D0–D7)	16 VCC
2 CP shift clock (rising edge)	15 /CE clock enable (active LOW, tie to GND)
3 D4 parallel input	14 D3 parallel input
4 D5 parallel input	13 D2 parallel input
5 D6 parallel input	12 D1 parallel input
6 D7 parallel input (first out)	11 D0 parallel input (last out)
7 /Q7 inverted serial output	10 DS serial data in (for daisy-chaining)
8 GND	9 Q7 serial output

Switches & output

Keypad 4×4

Sixteen buttons wired as a grid with only 8 pins: pressing a key connects its ROW pin to its COLUMN pin. To read it, make the column pins outputs (one LOW at a time) and the row pins INPUT_PULLUP, then see which row goes LOW. CLICK a key to press it; click it again to release.

#	Pin / terminal
1	row 1
2	row 2
3	row 3
4	row 4
5	column 1
6	column 2
7	column 3
8	column 4

Inspector settings: Key pressed

DIP switch (4 positions)

Four tiny independent slide switches in one package. Each connects its left pin to its right pin when ON. CLICK a slider to flip it. Handy for setting a binary number, an address or a configuration.

#	Pin / terminal
1	switch 1 left
2	switch 2 left
3	switch 3 left
4	switch 4 left
5	switch 1 right
6	switch 2 right
7	switch 3 right
8	switch 4 right

Inspector settings: Switch 1 ON, Switch 2 ON, Switch 3 ON, Switch 4 ON

Display & motion

Lamp (filament bulb)

A filament bulb. Its resistance is about 10 times lower when cold, so it draws a big “inrush” current at switch-on and then settles as the filament heats up and glows. Choose the rating that matches your supply: more than about 30 % over the rating burns it out.

#	Pin / terminal
1	terminal A
2	terminal B

Inspector settings: Rated voltage

LED bar graph (10 LEDs)

Ten LEDs in a row, as used in level meters. Anodes (A1–A10) are along the bottom, cathodes (K1–K10) along the top: to light segment n connect Kn to GND and give An a HIGH. With “built-in resistors” each segment already has 220 Ω; turn it off to add your own. Drive it from Arduino pins, from a CD4017 (running light), or from an analog level.

Pins (first half)	Pins (second half)
anode 1	cathode 1
anode 2	cathode 2
anode 3	cathode 3
anode 4	cathode 4
anode 5	cathode 5
anode 6	cathode 6
anode 7	cathode 7
anode 8	cathode 8
anode 9	cathode 9
anode 10	cathode 10

Inspector settings: Colour, Series resistors

8×8 LED matrix

Sixty-four LEDs in a grid. ROW pins are the anodes (+), COLUMN pins are the cathodes (−): the LED at (row, column) lights when its ROW is HIGH and its COLUMN is LOW. Light one row at a time very quickly (multiplexing) and your eye sees the whole picture. Each column has a built-in 220 Ω resistor. Write 64 pixels using only 16 pins.

Pins (first half)	Pins (second half)
row 1 (anode +)	column 1 (cathode −)
row 2 (anode +)	column 2 (cathode −)
row 3 (anode +)	column 3 (cathode −)
row 4 (anode +)	column 4 (cathode −)
row 5 (anode +)	column 5 (cathode −)
row 6 (anode +)	column 6 (cathode −)
row 7 (anode +)	column 7 (cathode −)
row 8 (anode +)	column 8 (cathode −)

Inspector settings: Colour

I²C LCD (16×2 / 20×4)

A character LCD with a small I²C “backpack”, so it needs only FOUR wires: GND, VCC (5 V), SDA to Arduino A4 and SCL to Arduino A5. Use the LiquidCrystal_I2C library: LiquidCrystal_I2C lcd(0x27, 16, 2); lcd.init(); lcd.backlight(); lcd.print("Hello"). The address in the sketch must match the module (0x27 or 0x3F). Without lcd.backlight() the text is too dark to read.

#	Pin / terminal
1	GND
2	VCC (5 V)
3	SDA (to Arduino A4)
4	SCL (to Arduino A5)

Inspector settings: Size, I²C address

OLED 128×64 (SSD1306, I²C)

A small, sharp black-and-white graphic screen with 128 × 64 pixels and four I²C wires: GND, VCC, SDA to Arduino A4 and SCL to A5. Use the Adafruit_SSD1306 library: Adafruit_SSD1306 display(128, 64, &Wire, -1); display.begin(SSD1306_SWITCHCAPVCC, 0x3C); then clearDisplay(), setTextSize(), setCursor(), println(), drawPixel / drawLine / drawRect / fillRect / drawCircle, and finally display() to show what you drew.

#	Pin / terminal
1	GND
2	VCC (3.3–5 V)
3	SCL (to Arduino A5)
4	SDA (to Arduino A4)

Stepper motor (28BYJ-48)

A motor that turns in tiny exact STEPS instead of spinning freely. Four coils (IN1–IN4) are energised in a sequence and the magnet rotor follows. Connect COM (the red wire) to 5 V and drive each coil end to ground, using a ULN2003 (a pin cannot supply 80 mA). Energise the coils in order 1-2-3-4 to turn one way, 4-3-2-1 for the other. A 2-coils-at-a-time sequence has more torque; one-at-a-time is simplest. One full turn of the shaft takes 4096 half-steps.

#	Pin / terminal
1	COM (red wire, to +5 V)
2	coil 1 (blue)
3	coil 2 (pink)
4	coil 3 (yellow)
5	coil 4 (orange)

Sensors

PIR motion sensor (HC-SR501)

Detects moving warm bodies (people). OUT goes HIGH (3.3 V) when it sees movement and stays HIGH for the “hold time” afterwards. It needs 5 V on VCC (4.5–20 V). CLICK the sensor to simulate somebody walking past (click again to stop). Connect OUT to an Arduino input.

#	Pin / terminal
1	VCC (4.5–20 V)
2	OUT (3.3 V when motion)
3	GND

Inspector settings: Motion in front, Hold time

Temperature sensor (LM35)

Analog temperature sensor with a very simple rule: output = 10 mV per °C (25 °C = 0.25 V). No offset to subtract: tempC = volts × 100. It needs 4–30 V on +Vs and, powered from a single supply, reads 2 °C and above. Use the slider to change the temperature.

#	Pin / terminal
1	+Vs (4–30 V)
2	Vout (10 mV / °C)
3	GND

Inspector settings: Temperature

Temperature & humidity (DHT11 / DHT22)

Measures temperature AND humidity and sends them over ONE data wire. Read it from the Arduino with the DHT library: DHT dht(2, DHT11); dht.begin(); then dht.readTemperature() (°C) and dht.readHumidity() (%). The DHT11 is whole numbers (0–50 °C, 20–90 %); the DHT22 is more accurate (one decimal, −40…80 °C, 0–100 %). The model in the sketch must match the sensor. Use the sliders to change the room. Wait 2 seconds between reads, as in real life.

#	Pin / terminal
1	VCC (3–5.5 V)
2	DATA (single wire to the Arduino)
3	GND

Inspector settings: Sensor model, Temperature, Humidity

Joystick module

Two potentiometers (left–right and up–down) plus a push button, like a game controller. VRx and VRy are analog voltages between 0 V and 5 V: about 2.5 V in the middle. SW is pulled to GND when you press the stick: use INPUT_PULLUP on that pin. Use the sliders for the position; CLICK the module to press the button.

#	Pin / terminal
1	GND
2	+5V
3	VRx (horizontal)
4	VRy (vertical)
5	SW (button, LOW when pressed)

Inspector settings: Left ← → Right, Up ↑ ↓ Down, Button pressed

Example circuits

7805 regulator: steady 5 V from a 12 V supply (Intermediate)

The 7805 turns 12 V into a steady 5 V for the LED. Select the battery and change its voltage in the inspector (try 8 V, then 5 V): the output stays at 5 V until the input drops below about 7 V. Select the regulator to see the heat it wastes.

555 + CD4017: LED running light (Advanced)

A 555 makes a slow clock (about 3 Hz); every pulse the CD4017 moves its single HIGH output on to the next of Q0–Q9, so a light runs along the LED bar graph. Change C to make it faster or slower.

Digital counter: 555 → 74HC161 → CD4511 → 7-segment (Advanced)

No microcontroller: a 555 clock drives a 4-bit binary counter, a BCD decoder turns the binary number into the segment pattern, and a 7-segment display shows it. It counts 0–9, then blanks for 10–15 (the decoder only knows decimal digits).

Arduino: stepper motor with a ULN2003 driver (Advanced)

Four pins energise the stepper coils one after another through a ULN2003 (a pin alone cannot supply 80 mA). Select the motor to watch the half-steps counted. Change the order in the sketch to reverse it, or shorten the delay to go faster.

Arduino: Stepper library, one turn each way (Intermediate)

The normal way to drive a stepper motor: the Stepper library does the coil sequence for you. step(2048) is one full turn of the 28BYJ-48 shaft; a negative number turns the other way. Select the motor to watch the steps.

Arduino: 4×4 keypad (Intermediate)

Click a key on the keypad (click again to release) and the Arduino prints it in the Serial tab. Eight wires read sixteen keys: the Keypad library scans rows and columns for you.

LM393 comparator: automatic night light (Intermediate)

Select the light sensor and drag its light slider down. When it gets dark the LDR voltage rises above the reference (set by two equal resistors) and the comparator output pulls LOW, which lights the LED.

555 one-shot timer (monostable) (Advanced)

Press and release the button: the 555 turns the LED on for exactly one time period (about 3 seconds, T = 1.1 × R × C) and then switches it off by itself. Change R or C to change the time.

SCR latching alarm (Advanced)

Press the GATE button once: the SCR switches on and the LED STAYS lit after you let go. That is latching. Open the RESET switch (click it) and close it again to clear the alarm, because the current must fall to zero for the SCR to let go.

Transformer: stepping AC voltage down (Intermediate)

A 10 V peak, 50 Hz source feeds the primary of a 2 : 1 transformer. The scope shows the secondary is a copy of the wave at HALF the size (5 V peak). Select the transformer and change the turns ratio to 1 : 2 to step the voltage UP: the current then falls. Replace the AC source with a battery to see that a transformer ignores DC.

8-bit ADC: a binary voltmeter (Intermediate)

Move the potentiometer (select it and use the slider): the ADC converts the wiper voltage into an 8-bit number and the LED bar graph shows its bits, D7 (128) on segment 1 down to D0 (1) on segment 8. At the start 3 V gives 153 = 10011001, so segments 1, 4, 5 and 8 light.

Arduino: DHT11 temperature and humidity (Intermediate)

The DHT library reads temperature and humidity over one wire. Select the sensor and move its sliders: the values appear in the Serial tab and the Plotter tab (printed as label:value pairs). The DHT11 gives whole numbers; a DHT22 would give decimals.

Arduino: I²C LCD with four wires (Beginner)

Only four wires: GND, VCC, SDA (to A4) and SCL (to A5). The LiquidCrystal_I2C library writes the text. Select the display to see its status; if the sketch address (0x27) does not match the module the screen explains why.

Arduino: OLED graphics (SSD1306) (Intermediate)

A 128 × 64 pixel graphic display on four I²C wires. The sketch writes text and draws a bar, a frame and a bouncing dot. Nothing appears until display.display() copies the drawing to the screen.